MA3521 — Eigenpair Solutions

Name:

Problem 1 Solve $\mathbf{y}' = A\mathbf{y}$, $\quad A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$, $\quad \mathbf{y}(0) = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$

1a. Find the eigenvalues and eigenvectors.

$\det(A - \lambda I) =$ $= 0$

$\lambda_1 =$ $\quad \mathbf{v}_1 =$

$\lambda_2 =$ $\quad \mathbf{v}_2 =$

1b. Write the GS and use the IC to find the constants.

$\mathbf{y}(t) =$

$V\mathbf{c} = \mathbf{y}(0)$ $\quad V =$ $\quad \mathbf{c} =$

1c. As $t \to \infty$, $\mathbf{y}(t) \sim$

Problem 2 Solve $\mathbf{y}' = A\mathbf{y}$, $\quad A = \begin{pmatrix} -1 & -2 \\ -2 & -1 \end{pmatrix}$, $\quad \mathbf{y}(0) = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$

2a. Find the eigenvalues and eigenvectors.

$\det(A - \lambda I) =$ $\qquad = 0$

$\lambda_1 =$ $\qquad \mathbf{v}_1 =$

$\lambda_2 =$ $\qquad \mathbf{v}_2 =$

2b. Write the GS and use the IC to find the constants.

$\mathbf{y}(t) =$

$V\mathbf{c} = \mathbf{y}(0) \qquad V =$ $\qquad \mathbf{c} =$

2c. As $t \to \infty$, $\mathbf{y}(t) \sim$